

Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation

Abstract

Large language models (LLMs) are increasingly being used to support scientific hypothesis generation, ranging from direct prompting and iterative self-refinement to autonomous, multi-agent scientific workflows. A central architectural question is whether hypothesis generation is better served by a single LLM operating through structured reasoning and critique, or by multiple LLM-based agents that generate, challenge, rank, and evolve candidate hypotheses. This paper reviews and compares these approaches with particular attention to scientific novelty, validity, diversity, factual grounding, computational cost, and experimental verifiability. Single-LLM systems can be surprisingly effective when equipped with iterative self-refinement, retrieval, structured prompting, and test-time computation. Multi-agent debate systems introduce explicit informational diversity and role specialization and have shown improvements on several reasoning tasks; however, general multi-agent debate does not consistently outperform simpler ensembles or self-consistency approaches, and debate can suffer from agreement collapse, sycophancy, correlated errors, and substantial inference overhead. Recent scientific systems—including AI Scientist, Virtual Lab, and Google's AI co-scientist—suggest that multi-agent architectures become more compelling when agents are connected to literature retrieval, computational tools, experimental validation, and human scientific oversight rather than relying on textual debate alone. The evidence therefore does not support a simple superiority claim for either paradigm. Instead, the most promising architecture is a hybrid in which a strong single model performs broad ideation and refinement, while selectively invoked heterogeneous agents provide adversarial critique, independent evidence retrieval, novelty assessment, and experimental prioritization. Future research should evaluate these systems on prospective, blinded, domain-expert and experimentally validated benchmarks rather than relying primarily on LLM-based scoring.

Keywords: large language models; scientific discovery; hypothesis generation; multi-agent systems; debate; scientific reasoning; AI scientist; test-time computation; novelty; experimental validation

1. Introduction

Scientific progress depends not only on collecting observations but also on constructing explanations that are sufficiently novel, plausible, and testable to motivate further investigation. The computational automation of this process predates modern LLMs: classical systems such as BACON attempted to discover scientific laws from structured observations, while literature-based discovery sought previously disconnected relationships across bodies of scientific knowledge [1,2]. Don Swanson's concept of “undiscovered public knowledge” is particularly relevant to contemporary LLM systems because it frames discovery as the identification and integration of relationships distributed across otherwise disconnected information sources [1].

LLMs have introduced a different paradigm. Rather than being designed solely around explicit symbolic discovery rules, they can synthesize textual and multimodal knowledge, generate candidate explanations, compare alternatives, summarize evidence, write experimental protocols, and iteratively revise their own outputs. Recent studies demonstrate that LLMs can generate hypotheses in data-driven settings [3], produce open-domain scientific hypotheses from raw web corpora [4], and generate research ideas that expert researchers sometimes rate as highly novel [5]. Zhou et al. demonstrated iterative hypothesis updating guided by a reward function, while Yang et al. constructed an open-domain scientific hypothesis-discovery benchmark explicitly targeting validity and novelty [3,4].

The emergence of agentic architectures has shifted the question from *what can one LLM generate?* to *how should multiple reasoning processes be organized?* Multi-agent debate (MAD) systems allow several model instances to independently propose solutions, critique one another, and converge on a final answer [6]. In parallel, single-model approaches such as Self-Refine and Reflexion show that substantial gains can be obtained without introducing multiple agents, simply by allocating additional inference steps to feedback, reflection, and revision [7,8].

This distinction is particularly consequential for scientific hypothesis generation. The objective is not merely to obtain the most likely answer to a closed benchmark question. A useful scientific hypothesis may be uncertain, surprising, difficult to evaluate automatically, or even inconsistent with the model's dominant prior expectations. Systems therefore must balance **novelty against plausibility**, **diversity against coherence**, and **creative exploration against evidential grounding**. Furthermore, the ultimate criterion is not linguistic quality but whether the hypothesis survives empirical scrutiny.

This paper compares single-LLM and multi-agent debate approaches for scientific hypothesis generation. It argues that neither architecture is inherently superior. Rather, their relative value depends on whether additional inference produces genuinely independent information, whether the system is externally grounded, and whether candidate hypotheses are evaluated through objective scientific criteria and experiments.

2. Background and Related Work

2.1 Scientific hypothesis generation as computational discovery

Classical computational discovery systems treated scientific reasoning as a structured search problem. BACON, for example, used heuristic procedures to identify regularities and formulate laws from numerical data [2]. Literature-based discovery introduced a complementary paradigm in which useful scientific relationships could emerge by connecting concepts that were not directly linked in existing publications [1].

Modern LLM-based hypothesis generation retains aspects of both paradigms. It can perform open-ended synthesis over literature, but it can also generate candidate mechanisms, explanations, and predictions from observations or datasets. Zhou et al. developed a method in which initial hypotheses generated from examples are iteratively updated using an exploration-exploitation objective inspired by multi-armed bandits [3]. Yang et al. extended the problem toward open-

domain scientific hypothesis discovery, using raw web corpora rather than manually selected observations and explicitly targeting novel and valid hypotheses [4].

A major difficulty is that scientific novelty is not equivalent to textual novelty. A statement may be absent from a model's retrieved literature while being scientifically trivial, false, or already known in another terminology. Conversely, a genuinely valuable hypothesis may be expressed in language that resembles prior work. This motivates evaluation procedures that separate novelty, plausibility, importance, and testability.

2.2 Single-LLM reasoning and iterative refinement

A single-LLM system need not mean one-shot prompting. Self-Refine demonstrated that one model can act successively as generator, critic, and reviser, producing substantial improvements across multiple tasks [7]. Reflexion similarly uses linguistic feedback and episodic memory to improve subsequent decisions without changing model parameters [8]. These approaches show that the distinction between “single-agent” and “multi-agent” is partly architectural rather than computational: a single underlying model can execute multiple reasoning roles over time.

For hypothesis generation, a single model can therefore implement a pipeline such as: generate several candidate hypotheses, retrieve evidence, identify contradictions, estimate novelty, revise the candidates, and select experiments. The principal advantages are lower coordination complexity and consistent context management. The principal limitation is that successive reasoning stages may inherit the same latent assumptions and systematic errors.

This issue is important because LLM self-evaluation is imperfect. Si et al.'s large-scale study involving more than 100 NLP researchers found that LLM-generated research ideas were judged more novel than human ideas in their experimental setting but somewhat weaker in feasibility; the study also identified failures in LLM self-evaluation and a lack of diversity among generated ideas [5].

2.3 Multi-agent debate

Multi-agent debate makes disagreement an explicit computational resource. In the framework of Du et al., multiple LLM instances propose answers and reasoning processes and then debate them over multiple rounds before producing a consensus [6]. Their experiments reported improvements in mathematical and strategic reasoning and reductions in some forms of factual error [6].

The rationale is analogous to scientific peer review: independent proposals may expose assumptions, identify errors, or uncover explanations that a single reasoning trajectory fails to consider. Role-specialized systems generalize this approach. MetaGPT, for example, assigns different agent roles within structured workflows rather than simply asking multiple instances to chat freely [9]. Such architectures suggest that the critical factor may not be “more agents” per se, but **division of epistemic labor**.

However, multi-agent debate has not consistently produced superior results. Smit et al. found that existing debate protocols did not reliably outperform self-consistency or multiple independent reasoning paths, while performance depended substantially on protocol and hyperparameter choices [10]. Their results imply that some apparent benefits of debate may arise from aggregation rather than genuine interactive reasoning.

3. Single-LLM Versus Multi-Agent Debate for Scientific Hypothesis Generation

3.1 The single-LLM paradigm

A strong single-LLM hypothesis generator can be represented as

Research goal → **evidence retrieval** → **candidate generation** → **self-critique** → **refinement** → **ranking** → **experimental proposal**.

The model may generate many hypotheses internally or explicitly, but all stages share the same underlying model representation. This has several advantages.

First, the system is comparatively simple. There is no requirement to synchronize agents, maintain multiple conversational states, or determine how disagreement should be resolved. Second, the approach is computationally economical relative to repeated multi-agent interaction. Third, a unified model can maintain a consistent global representation of the research problem.

Evidence from Self-Refine suggests that iterative inference itself can provide meaningful gains over one-shot generation [7]. Similarly, Zhou et al. showed that iterative hypothesis refinement can outperform few-shot prompting on several predictive tasks [3].

The principal scientific weakness is **correlated reasoning failure**. If the model initially adopts an incorrect causal assumption, later self-critique may simply rationalize the assumption rather than overturn it. The problem becomes more serious when the model is asked to evaluate its own novelty or correctness. Si et al. explicitly observed limitations in LLM self-evaluation and idea diversity [5].

Nevertheless, single-LLM systems can substantially mitigate these weaknesses through external evidence. PaperQA2, for example, uses agentic literature retrieval and citation-oriented synthesis; its evaluation found performance competitive with or exceeding subject-matter experts on several scientific literature tasks [11]. This suggests that the scientific value of an LLM can be enhanced more effectively by improving **evidence access** than merely by multiplying conversational agents.

3.2 The multi-agent debate paradigm

A multi-agent hypothesis system can instead be expressed as

Research goal → **independent hypothesis generation** → **adversarial debate** → **evidence retrieval** → **hypothesis ranking** → **evolution/refinement** → **experimental prioritization**.

Multiple agents can be given different roles, such as:

1. hypothesis generator;
2. mechanism-focused critic;
3. literature verifier;
4. novelty assessor;
5. falsification specialist;
6. experimental designer;
7. meta-reviewer.

This structure introduces a form of epistemic diversification. A mechanistic critic may identify a biological inconsistency that a general generator misses; a literature agent may discover an apparently novel hypothesis already described under a different terminology; and an experimental agent may reject an attractive explanation because it cannot be tested economically.

Google's AI co-scientist represents one of the most explicit implementations of this paradigm. The system employs specialized Generation, Reflection, Ranking, Evolution, Proximity, and Meta-review agents and uses scientific debate and tournament-based comparison to iteratively improve hypotheses [12]. The system has also been evaluated in biomedical settings involving drug repurposing, liver-fibrosis targets, and bacterial gene-transfer mechanisms, including experimental validation of selected hypotheses [12].

3.3 Diversity is useful only when it is real

The theoretical advantage of multi-agent systems depends critically on **non-redundancy**. Five agents initialized with the same model, prompt, knowledge base, and objective may produce superficially different text but highly correlated reasoning. In such a case, increasing agent count may add computational cost without proportionally increasing epistemic coverage.

Smit et al.'s comparison of debate, self-consistency, and ensemble approaches is important here: debate was not consistently superior, indicating that simply creating conversational interaction does not guarantee better reasoning [10].

More recent work also identifies **sycophancy** as a serious multi-agent failure mode. Pitre et al. found that agents can reinforce one another rather than critically engage, increasing the number of debate rounds while reducing efficiency and reliability [13].

For scientific discovery, premature agreement may be especially damaging. Scientific breakthroughs can depend on hypotheses that appear implausible before evidence is collected. A debate system optimized for consensus may eliminate unusual hypotheses too early.

3.4 The importance of external validation

The strongest scientific systems increasingly combine LLM reasoning with external tools rather than relying solely on language-based debate. Coscientist, for example, integrates GPT-4 with web search, documentation retrieval, code execution, and laboratory automation to design and perform chemical experiments [14]. The Virtual Lab similarly uses a principal-investigator agent and specialized scientific agents, with human supervision, to address interdisciplinary problems such as designing SARS-CoV-2 nanobody binders [15].

The implication is important: **multi-agent debate should not be treated as an end in itself**. Its scientific value is greatest when disagreement results in new evidence, new computations, alternative experiments, or more robust falsification tests.

4. Current Approaches and Developments

4.1 Iterative test-time computation

A major current trend is scaling reasoning at inference time rather than relying exclusively on larger pretrained models. Single-agent systems can use repeated sampling, self-critique, tree-based exploration, or reflective memory [7,8]. Multi-agent systems use the same principle through parallel agents, debate rounds, tournaments, and evolutionary selection [6,12].

The AI co-scientist explicitly frames scientific reasoning as test-time compute scaling and uses asynchronous multi-agent execution, self-play debate, tournament ranking, and hypothesis evolution [12].

4.2 Autonomous scientific research systems

The AI Scientist is a complementary development in which LLM-based agents generate research ideas, write code, execute experiments, visualize results, produce manuscripts, and perform automated review [16]. The 2026 end-to-end version extends this architecture toward more autonomous scientific research and reports an end-to-end pipeline spanning idea generation through manuscript creation and peer review [17].

AI Scientist-v2 further introduced progressive agentic tree search and an experiment-manager agent, reporting an entirely AI-generated paper that passed the first round of peer review at an ICLR workshop [18]. These results are notable, but they should not be interpreted as evidence that autonomous systems can yet replace expert scientific communities. Workshop acceptance and automated evaluation remain substantially weaker endpoints than independent experimental confirmation and durable scientific adoption.

4.3 Literature-grounded hypothesis generation

Scientific hypothesis generation requires access to an evolving literature base. Systems such as PaperQA2 demonstrate that retrieval-enhanced agents can substantially improve scientific information synthesis and contradiction detection [11]. The challenge is therefore shifting from merely generating fluent hypotheses to constructing **evidence-linked hypotheses** whose claims can be traced to identifiable observations.

This direction is particularly compatible with multi-agent architectures. One agent can propose the hypothesis, another can search for supporting evidence, another for contradictory evidence, and another can audit whether the final formulation is genuinely novel. The architecture then resembles scientific peer review more closely than conventional chatbot interaction.

5. Comparative Analysis

Criterion	Single-LLM with structured reasoning	Multi-agent debate
Generation diversity	Moderate; can be increased by sampling	Potentially high
Context consistency	High	Lower as agent interactions expand

Criterion	Single-LLM with structured reasoning	Multi-agent debate
Coordination overhead	Low	High
Computational cost	Lower	Often substantially higher
Error correction	Self-critique and iterative refinement	Cross-agent critique and debate
Correlated errors	High risk	Reduced only with genuine heterogeneity
Sycophancy/consensus collapse	Less prominent	Significant risk
Role specialization	Usually simulated by prompting	Naturally supported
Evidence retrieval	Straightforward	Can be distributed across agents
Novelty assessment	Vulnerable to self-evaluation bias	Multiple evaluators possible
Experimental planning	Strong when tools are integrated	Strong when specialized agents are integrated
Human oversight	Simple to implement	More complex but potentially richer
Best scientific use	Fast ideation and iterative refinement	Complex, interdisciplinary, evidence-intensive discovery

The central comparison is therefore not “one model versus many models,” but **one reasoning trajectory versus multiple partially independent reasoning trajectories**. A single model with ten sequential refinement steps may consume more computation than a nominally “multi-agent” system with two agents and one round. Conversely, ten identical agents may provide less diversity than one model prompted to explore distinct mechanisms.

6. Research Challenges and Limitations

6.1 Defining and measuring scientific novelty

Novelty is arguably the most difficult evaluation criterion. A hypothesis can be linguistically novel but conceptually redundant. Conversely, a hypothesis that appears similar to previous work may extend that work into a genuinely new domain. Existing evaluations therefore risk conflating novelty with unfamiliar wording or absence from a search corpus.

Si et al. show that expert assessments of novelty and feasibility can diverge [5], while Yang et al. demonstrate the difficulty of constructing open-domain benchmarks in which novelty is explicitly part of the task definition [4].

Future evaluation should separate at least four dimensions: **literature novelty, conceptual novelty, empirical plausibility, and downstream scientific utility**.

6.2 Hallucination and false evidence

Scientific hypothesis generation is unusually sensitive to hallucination because a fabricated citation, nonexistent result, or incorrectly stated mechanism can make a hypothesis appear far more plausible than it is. Retrieval-based systems reduce this risk but do not eliminate it.

Recent work on agentic scientific synthesis has shown that even the evaluation of citation faithfulness can itself be unreliable, with verifier choice materially affecting unsupported-citation estimates [19].

This creates a recursive verification problem: a model cannot reliably establish that its own evidence is correct merely by generating another model-based verification step. Independent databases, source documents, structured knowledge representations, and human review remain important.

6.3 Correlated errors in multi-agent systems

Using multiple agents does not automatically create independence. If all agents share the same model, literature corpus, prompt structure, and latent biases, their errors may be strongly correlated. Debate can then produce **consensus without correctness**.

The emerging literature on sycophancy and disagreement collapse strengthens this concern [13]. Multi-agent systems should therefore be evaluated using measures of information diversity, not merely final-answer accuracy.

6.4 Cost, latency, and scalability

Multi-agent debate multiplies model calls and communication. Smit et al. explicitly evaluated trade-offs among accuracy, cost, and time, while subsequent work on economical multi-agent communication has highlighted substantial token overhead in multi-agent pipelines [10,20].

For scientific applications, cost matters because meaningful hypothesis generation may require thousands of candidate ideas, repeated literature searches, simulations, and experimental proposals. A system that improves quality by 5% while increasing inference cost by an order of magnitude may be unattractive outside high-value research domains.

6.5 Scientific validation remains the bottleneck

LLMs can generate hypotheses quickly, but scientific knowledge advances only when hypotheses can be tested. The strongest recent studies therefore combine model generation with experiments. The breast-cancer study by Zheng et al. evaluated GPT-4-generated drug-combination hypotheses experimentally and found several combinations with positive synergy relative to controls [21]. The AI co-scientist similarly reports experimental validation of selected biomedical hypotheses [12].

These studies also demonstrate the limit of purely computational benchmarks: a hypothesis-generation system can score highly according to human or LLM judges without producing a durable scientific discovery.

6.6 Safety and dual-use risk

As systems become capable of generating experimental protocols and connecting reasoning agents to laboratory infrastructure, the risk profile changes. Coscientist demonstrated semi-autonomous experimental execution in chemistry [14], while recent work has proposed dedicated safety monitoring architectures for scientific agents [22].

Scientific hypothesis-generation systems should therefore treat safety assessment as part of the architecture rather than an after-the-fact filter.

7. Research Gaps and Future Directions

7.1 Controlled head-to-head scientific evaluations

The literature currently contains many demonstrations of either single-agent or multi-agent capabilities, but comparatively few controlled experiments that isolate architecture as the independent variable. A rigorous benchmark should hold constant the underlying model, retrieved evidence, computational budget, and output format while comparing:

1. one-shot generation;
2. single-LLM iterative refinement;
3. independent multi-sampling;
4. multi-agent debate;
5. heterogeneous multi-agent debate;
6. hybrid debate-plus-tool systems.

Such experiments should be conducted across several scientific disciplines rather than only general reasoning benchmarks.

7.2 Compute-matched evaluation

A major methodological gap is failure to compare architectures under equal inference budgets. “More agents” frequently means “more computation.” The relevant question is therefore not whether a 10-agent system outperforms a single call, but whether it outperforms a single agent receiving the same computational budget.

This is especially important because current evidence suggests that additional test-time computation itself improves performance [12]. A fair comparison should report quality as a function of token usage, wall-clock time, number of model calls, and monetary cost.

7.3 Evidence-first debate

Future systems should shift from **argument-first debate** to **evidence-first debate**. Instead of asking two agents merely to argue for competing hypotheses, the system should require each agent to retrieve independently verifiable evidence, identify falsifiers, quantify uncertainty, and specify what observation would change its conclusion.

A useful protocol could be:

Generate → **independently retrieve** → **challenge** → **falsify** → **simulate** → **rank** → **experimentally test**.

This would make the debate computationally closer to scientific peer review.

7.4 Heterogeneous agent populations

Architectures should deliberately introduce epistemic heterogeneity. Instead of ten copies of the same model, agents could differ in model family, retrieval corpus, scientific discipline, reasoning strategy, or objective function.

For example, one agent could optimize explanatory simplicity, another mechanistic plausibility, another novelty, another experimental feasibility, and another falsifiability. The purpose would not be to manufacture disagreement but to increase the probability that important weaknesses are detected.

7.5 Human scientists as high-level arbiters

Current evidence favors scientist-in-the-loop architectures over unrestricted autonomous discovery. Google's AI co-scientist explicitly positions the scientist as part of the workflow, allowing human researchers to specify goals, constraints, and feedback [12].

The most effective role for humans may therefore be **problem selection, constraint setting, experimental prioritization, and final epistemic judgment**, while LLM systems perform large-scale search and synthesis.

7.6 Experimentally validated benchmarks

The ultimate benchmark should follow hypotheses longitudinally. A useful evaluation would generate hypotheses from literature available before a specified cutoff date, then determine whether later publications or experiments independently confirm, reject, or modify those hypotheses.

Such temporal evaluation would reduce contamination and directly measure whether an AI-generated hypothesis contributes useful scientific knowledge rather than merely reproducing existing literature.

7.7 Provenance and reproducibility

Scientific AI systems should preserve complete provenance: source documents, retrieval queries, model versions, prompts, candidate hypotheses, debate transcripts, discarded alternatives, and selection criteria. This is essential because a final hypothesis may depend on a chain of intermediate reasoning that is difficult to reconstruct from the output alone.

Future benchmarks should therefore evaluate not only *what hypothesis was produced*, but also whether the system can provide a reproducible evidence trail supporting how the hypothesis was constructed.

8. Conclusion

The available evidence does not establish that multi-agent debate systems are universally superior to single-LLM systems for scientific hypothesis generation. Single-LLM approaches can achieve substantial improvements through iterative refinement, reflection, retrieval, and additional test-time computation [3,7,8]. Multi-agent systems provide a complementary mechanism for generating diversity, specialization, adversarial critique, and structured comparison, and recent scientific

systems demonstrate their potential when coupled to literature retrieval, computational tools, experimental validation, and human oversight [12,14–18].

The key distinction is therefore not the number of agents but the **independence, specialization, and evidential value of their reasoning**. Multi-agent debate is most promising when agents contribute information that a single reasoning trajectory is unlikely to produce. When agents are homogeneous and merely persuade one another, the additional computation can produce little benefit and may amplify sycophancy or consensus errors [10,13].

For scientific discovery, the strongest near-term architecture is likely hybrid: a capable single LLM generates a broad hypothesis space and performs inexpensive iterative refinement, while selectively invoked heterogeneous agents independently verify literature, challenge assumptions, assess novelty, propose falsification tests, and rank experimental candidates. The final arbiter should increasingly be external evidence and experiment rather than model consensus.

Consequently, future research should move beyond the question “**Which architecture wins?**” toward the more scientifically meaningful question “**Under what conditions does additional reasoning produce new, valid, testable knowledge per unit of computation?**” Answering that question will require controlled, compute-matched, prospective, domain-diverse, and experimentally grounded evaluations. Until such evidence becomes widespread, multi-agent debate should be regarded as a powerful architectural option—not as an established replacement for carefully designed single-LLM hypothesis-generation pipelines.

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